Probability

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Probability

- Random variable . . . X
- Instantiation of the random variable. . . x
- Probability that value x is observed. . . $P(x)$
- Normalisation condition $\sum_x P(x) = 1$
- If x is a continuous variable, then the normalisation condition becomes: $\int_{min(X)}^{max(X)} P(x)dx = 1$

Examples

- Toss a coin
  - $$P(x = \text{head}) = P(x = \text{tail}) = \frac{1}{2} \tag{1}$$
  - $$\sum_{x=\text{head,tail}} P(x) = \frac{1}{2} + \frac{1}{2} = 1 \tag{2}$$
- How about throwing a dice?

Markov chains

Major distinction:

- Continuous time: $T = 0.2313, 0.345, 0.50284, \ldots$
- Discrete time: $T = 1, 2, 3, \ldots$

Properties:

- A set of states $\{s_1, s_2, \ldots, s_N\}$
- Transitions between states, i.e. $s_{65} \rightarrow s_{92}$
- Transition probabilities/rates that quantify the transition.
- **The transition rate/probability only depends on the current state, but not on the history of states.**

Continuous time vs. discrete time

- Transitions in discrete time MCs are probabilities. Outgoing probabilities must sum to one, for each node.
- Transitions for CTMCs (continuous time Markov chains) are rates. They include time as well.
  - Assume from sum state s_i there are three possible transitions, with rates k_1, k_2, k_3.
  - The probability that the first transition happens is $\frac{k_1}{k_1+k_2+k_3}$.
  - A rate (roughly) means the average number of transitions that happen within a unit time. (You need to choose the unit time yourself).
  - The average waiting time for the first transition is $\frac{1}{k_1+k_2+k_3}$.

Initial condition and steady state behaviour

- A Markov chain is usually started in some initial state, say s_{23}, i.e. $P_{T=0}(s_{23}) = 1$.
- One is usually interested in the probability of states at some later time T.
- In the long run, technically at time $T = \infty$, Markov chains approach a "steady state," characterised by a steady state probability $P_{T=\infty}(s_i)$ for each of the possible states s_i.
- Under (mild) conditions, these steady state probability distribution will not depend on the initial state.
- One way to express this is to say that the system has "forgotten" its initial state.

Discussion question

1. We said that the transition probability does not depend on the history of the Markov chains, only on the current state. Yet, on the last slide I said that in steady state the system has forgotten its past....How can it forget its past when it never remembered it in the first place?
2. Assume a random walker on a 2×2 grid. The walker is allowed to go up, down, left and right. Represent this as a MC model.
3. Often one is interested in the long-term behaviour of a Markov chain, i.e. the steady state probabilities. Now, imagine the inverse problem. If you specify a desired long-term probability distribution, and a transition graph, how should you specify the transition probabilities, that achieve this?

Further reading

- Check my book:
  Barnes, D. and Chu, D. (2015). Guide to Simulation and Modeling for Biosciences. [Online]. Springer. Available at: http://dx.doi.org/10.1007/978-1-4471-6762-4.
  It has an entire chapter devoted to MCs.
- Check the "Prism model checker" software.
- Even better: Program your own Markov chain models using your favourite language.